

Shih-Yi (Matteo) Soo

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SUMMARY

Senior Machine Learning Engineer with 5+ years building NLP and GenAI systems in finance. Took Taiwan's first financial conversational AI from PoC to production; now tech lead & TPM for agentic applications — spanning RAG, agent orchestration (Google ADK), and GCP infrastructure.

WORK EXPERIENCE

E.Sun Financial Holding Co., Ltd

Senior Machine Learning Engineer

Taipei, Taiwan

Mar 2023 – Present

- **Agentic Wealth Management Chatbot:** Served as project tech lead and TPM (timeline, requirement alignment); shipped an internal agentic chatbot in 3 months (Q2 2026), serving **800+ wealth management RMs** over a 2,000+ page enterprise knowledge base.
- **Architecture:** Architected on Google ADK with modular workflow skills; adaptive retrieval and query reformulation **cut retrieval misses by 20%** (hit rate 90% → 92%), with deterministic guardrails (retrieval-count cap via `tool_context` state) preventing runaway loops.
- **RAG Chatbot & Cloud Infrastructure:** Led Taiwan's first conversational GenAI service in finance (PoC → in-app production launch July 2025); built a dual-track system (deterministic enforcement for compliance decisions, RAG for market Q&A) with hybrid retrieval, VLM-based ETL, and automated SLA alerting; deployed on GCP (Cloud Run, AlloyDB). Demoed at **NVIDIA GTC 2025 & FinTech Taipei 2024**; featured in 50+ media outlets.
- **AI Community:** Organized Taiwan's first RAG Hackathon (487 teams); obtained utility model and invention patents for a GPT-based automated evaluation methodology.

Machine Learning Engineer

Jul 2021 – Mar 2023

- **Speech (TTS):** Built in-house TTS service (RTF 0.3) using Tacotron2/VITS with 20-hour custom dataset; fine-tuned IPA-based models to support 4-language interleaving for telephony IVR systems.
- **BERT Chatbot & On-premise Infrastructure:** Managed 12+ BERT-based chatbots on Kubernetes; deployed gateway APIs handling 1,000+ monthly calls.

Walsin Lihwa Corporation

Software Engineer - Internship

Taipei, Taiwan

Apr 2018 – Jul 2018

- Applied PageRank and bi-LSTM for industrial text summarization; automated HR web scraping for talent acquisition.

TECHNICAL SKILLS

Generative AI	LLM, RAG, Agentic Orchestration, Google ADK, PyTorch
NLP & Speech	BERT, Text-to-Speech (TTS), Text Mining, IPA Encoding
Cloud & Infra	GCP (Cloud Run, AlloyDB, BigQuery, Vertex AI), Kubernetes, Docker, Airflow
Dev Tools	Python, FastAPI, Git, SQL, Linux

EDUCATION

National Tsing Hua University

M.S. in Computer Science

2018 – 2021

Hsinchu, Taiwan

Soochow University

B.S. in Computer Science and Information Management

2014 – 2018

Taipei, Taiwan